TPC-417: A Finite Full-Operator Bound for the Four-Shell C1 Proxy

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Abstract

We extend the four-shell pooled C1 proxy from one adjacent entry to its full finite window matrix. For four complete prime shells and each of $H \in \{16, 32, 66, 128\}$, we derive the exact endpoint/interior diagonal energies and a block decomposition of the locally normalized matrix. A finite, explicit bound is obtained: $\|Z\|_2 \le 2/(a_{\min}\sqrt{H}) + 16|A|/V_-$. The certificate and an independent aggregate replay use exact rational arithmetic and retain all 75483 primes. This is a finite synthetic-operator theorem only: it does not provide a growing bound, physical coefficient identification, arithmetic L^2 saving, Route-B closure, or a twin-prime result.

1 Finite model

Let $I_o = \{o, \dots, o+N-1\}$, $N = 4H$, and $T_d = H^2/(H^2+d^2)$. Pool the complete shells $Q < p \le 2Q$ for $Q = 65536, 131072, 262144, 524288$, with counts $5709, 10749, 20390, 38635$. Order the resulting primes and set $a_i = p_i^3/[Q_i^2(p_i - 1)]$. Even indices divide o and odd indices divide $o + N$; the CRT gives such an $o > B$ for every B. Put $P_- = \sum_{i \text{ odd}} a_i$, $P_+ = \sum_{i \text{ even}} a_i$, $A = P_+ - P_-$, and $V_\pm = \sum_{i \text{ of sign} \pm} a_i^2$.

2 Exact matrix identities

For $0 \le r, s < N$ and $r \ne s$, the diagonal-deletion identity gives

$$M_{rs} = T_{r-s}\{-A + b_r + b_s\}, \qquad b_0 = P_+, \quad b_r = 0 \ (r \ge 1).$$

Define $S_r = \sum_{s \ne r} T_{r-s}^2$. The literal masks therefore give

$$D_0 = V_- S_0, \qquad D_r = V_- S_r + V_+(S_r - T_r^2) \quad (1 \le r < N).$$

After conjugation by $D^{-1/2}$, the matrix has the block form

$$Z = \begin{pmatrix} 0 & q^T \\ q & C \end{pmatrix}, \quad q_r = \frac{P_- T_r}{\sqrt{D_0 D_r}}, \quad C_{rs} = \frac{-A T_{r-s}}{\sqrt{D_r D_s}} \quad (r \ne s), \qquad C_{rr} = 0.$$

3 Bound

At least H distances on one side of every interior point are at most H, so $S_r \geq H/4$. Cauchy–Schwarz gives $P_-^2 \leq m_- V_-$ and $V_- \geq m_- a_{\min}^2$. Since $\sum_{r=1}^{N-1} T_r^2 \leq S_0$,

$$\|q\|_2^2 \leq \frac{4P_-^2}{V_-^2 H} \leq \frac{4}{a_{\min}^2 H}.$$

Also $D_r \geq V_- H/4$. Splitting the kernel sum at H gives $\sum_{d \geq 1} T_d \leq 2H$ on each side, hence every absolute row sum of C is at most $16|A|/V_-$. Symmetry gives the same 2-norm bound. The triangle inequality applied to the star block and the bulk block proves

$$\boxed{\|Z\|_2 \leq \frac{2}{a_{\min}\sqrt{H}} + \frac{16|A|}{V_-}}.$$

The star certificate records the exact Cauchy–Schwarz envelope, not an exact spectral norm. All bound coefficients are exact rational quantities; no decimal observation enters the proof.

4 Scope and audit

The four heights are independently recomputed from exact fractions. The certificate records shell hashes, all aggregate quantities, positivity of the local diagonals, and both components of the bound. The producer, independent replay, and mutation checker are run in normal and optimized modes. The result is a full finite matrix bound for the declared synthetic proxy, not a bound uniform in growing H, Q, N and not a statement about the physical source signs.

finite full matrix bound	`PROVED_EXACT_FINITE`
growing operator theorem	`OPEN`
physical h_0/arithmetic sign	`OPEN`
arithmetic advance / fixed-power credit	`NO` / 0
Route-B / twin-prime result	`OPEN` / `NONE`

Reproduction

The project README lists the producer, independent replay, stress checker, and Bridge-B commands. The exact JSON certificate and proof package are part of the release.